AD DS FOUNDATIONS

One directory. Central management.

1 Topology

The physical or logical layout of a network.



2 Workgroup

Each computer manages its own local accounts.



3 Domain

Centralized identity and administration for joined computers.



4 AD DS = Active Directory Domain Services.

Stores and organizes users, computers and groups.



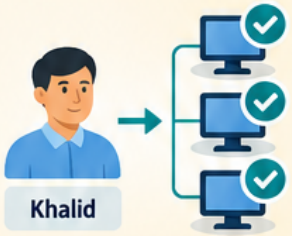
5 Domain Controller (DC)

A server running AD DS that holds directory data and authenticates domain identities.



6 Example

Create Khalid's domain account once. Use it on domain computers where logon is allowed.



7 Key Distinctions

Authentication: Who are you?

Authorization: What may you do?

A domain account does not automatically grant access to every server or folder.

DATABASE & PARTITIONS

What AD stores — and where

1 Schema



Defines object types and attributes.

Example:
user objects can have a telephone number.

2 Configuration



Forest-wide directory structure, sites, subnets and replication topology.

3 Domain



Directory objects for a domain, such as users, groups and computers.

4 Application



Service-specific directory data.

Example:
AD-integrated DNS records.

Replicates to selected directory replicas.

5 NTDS.DIT



AD directory database.

Default:
`C:\Windows\NTDS\ntds.dit`

6 SYSVOL



Group Policy files and logon scripts.

Default:
`C:\Windows\SYSVOL`

7 Transaction logs



Record database changes and support crash recovery.

Default locations can be changed during setup.
Transaction logs are not user logon history.

KERBEROS AUTHENTICATION

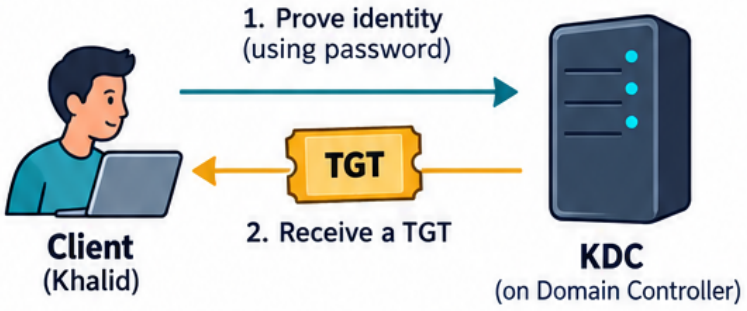
Identity through tickets

KDC = Key Distribution Center.
A service on a Domain Controller.

1 Sign in

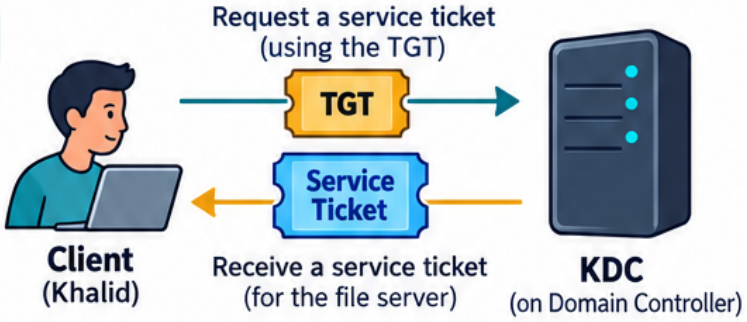
The client proves the user's identity to the KDC and receives a TGT.

TGT = Ticket Granting Ticket.



2 Request a service

The client uses the TGT to request a service ticket for the file server.



3 Access the resource

The client presents the service ticket. The file server validates it and checks permissions.



Example:

Read permission: Khalid can read the file.



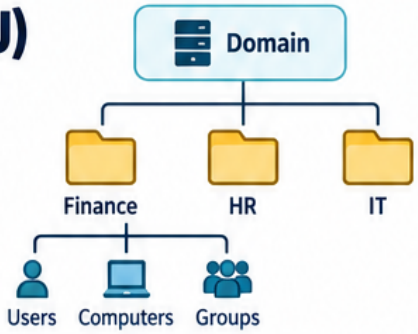
Read – Allowed

ORGANIZE, CONTROL & GROW

OU • Groups • Policies • Delegation

1 Organizational Unit (OU)

A directory container for users, computers, groups and other OUs. Organizes objects, supports Group Policy links and delegated administration.



- Organizes objects
- Supports Group Policy links
- Enables delegated administration

2 Security group

Collects members so resource permissions can be assigned to the group.
A user can belong to several groups.



Example

- Add users to a group
- Assign permissions to the group
- Users can be members of several groups

3 Policy vs permission

Policy

Configures behavior, such as automatic screen locking.

Permission

Allows an action, such as reading a folder.

Important

GPO settings apply to users or computers according to setting type, scope and filtering.

Groups do not receive GPO settings directly.

4 Scalability

Support more users, computers and directory workload as the organization grows.



Example

- More users
- More computers
- Higher directory workload
- Grows with organizational needs

5 Delegation

Give limited administrative rights for a specific task. Example: help desk can reset passwords for users in the Finance OU without becoming Domain Admins.



Department transfer: remove old group membership and add the appropriate new membership. Access changes may require a refreshed sign-in token.